

Plan 5 — LLM Cost Math + Business Model

Status: Locked 2026-04-24 **Companion docs:** [plan-1-auth.md](#), [plan-2-retention.md](#), [plan-3-integration-scope.md](#), [plan-4-integration-tech.md](#), [product-validation.md](#)

Executive summary

Skillnex defaults to **Claude Haiku 4.5** for per-competency narrative generation — the Day 9 review proved 94% safety-guard pass rate at 3× lower cost than Sonnet. Sonnet 4 is the upgrade tier, triggered automatically for outliers and senior employees, on-demand via a "Regenerate with deeper analysis" button, and for the cycle-close three-findings summary. Per-employee per-cycle cost is roughly \$0.022 (Haiku-only) to \$0.027 (hybrid). At a 500-employee mid-market customer paying \$7/seat/month, annual revenue is \$42,000 against ~\$260 in direct costs — gross margin north of 99%. Connector APIs (Salesforce, BambooHR, GitHub, etc.) are free for both customer and Skillnex; Workday Partner program at \$20-50K/year is the only meaningful integration cost and only kicks in when a paying enterprise demands Workday.

Locked decisions

#	Decision
1	Default model: Haiku 4.5 (proven 94% safe in Day 9 real-API review)
2	Sonnet upgrade trigger: all three — on-demand "regenerate" button + automatic for OUTLIER/REVIEW status + automatic for Director+ levels
3	Three-findings cycle-close summary: Sonnet (one big synthesis call per company per cycle)
4	Pricing model: per-seat at \$7/mo blended for pilot , volume discount at 1,000+ seats
5	Pilot pricing: free 90-day pilot , then \$7/seat/mo month-to-month for 6 months, then annual
6	Free tier: none in MVP ; reconsider at month 6 if growth needs it
7	Per-customer LLM budget cap: none ; soft alert internally if a single customer crosses \$100/cycle
8	Anthropic ZDR uplift: eat it as differentiator, don't pass through
9	Cost monitoring tooling: day one — <code>usage_log</code> table, dashboard in <code>/app/settings/billing</code>
10	Multi-tenant prompt cache amortization: enjoy it as margin gain , no surfaced pricing logic

Model selection

Model	Strengths	Weaknesses	\$/M input	\$/M output	Used for
Claude Haiku 4.5	Sub-second responses, structured-output reliable, 94% safety guard pass	Less rich prose, occasionally repetitive	\$1	\$5	Default — every per-competency narrative
Claude Sonnet 4	Better prose, more nuanced anomaly framing, smarter at edge cases	3× cost, 3× slower	\$3	\$15	Upgrade tier — outlier reviews, on-demand regeneration, cycle-close summary, Director+ employees
Claude Opus 4	Best prose	15× Haiku cost	\$15	\$75	Not used — bad price/value for templated narratives
GPT-4o	Comparable to Sonnet	Slightly cheaper input	\$2.50	\$10	Not used — already on Anthropic via DPA + ZDR roadmap

Hybrid rule in code:

```
``ts // lib/llm/router.ts function routeToModel(employee: EmployeeRecord, request: AnalyzeRequest): Model { if (request.regenerate_with_deeper_analysis) return MODEL_SONNET; if (employee.level && ["Director", "VP", "C-Level"].includes(employee.level)) return MODEL_SONNET; if (employee.flags.includes("score-vs-rating-mismatch") || employee.flags.includes("low-roi-senior")) { return MODEL_SONNET; } return MODEL_HAIKU; } ``
```

Cost equation

For one customer, one quarterly review cycle, the LLM cost is:

```
`` Cost = (employees × competencies × cost_per_narrative) + (employees × cost_per_generic_comment_check) + (1 × cost_per_three_findings_summary) + (regeneration_buffer × 1.3) // managers regen ~30% of drafts ``
```

For the per-competency model from the Before-After PDF:

- 5 competencies per employee
- ~2,500 system-prompt tokens (cached, ~95% hit rate across the run)
- ~800 user-message tokens per call (employee context for that competency)

- ~350 output tokens per narrative
- Generic-comment classifier: ~400 input + 50 output per detection × 5 competencies
- Three-findings synthesis: ~5,000 input + 1,200 output, one call per cycle

Per-employee per-cycle math (Haiku default)

Component	Tokens	Rate	Cost
System prompt cache write (one-time per cycle)	3,125	\$1/M	\$0.003
System prompt cache reads (4 of 5 competencies)	10,000	\$0.10/M	\$0.001
User input (5 calls × 800)	4,000	\$1/M	\$0.004
Narrative output (5 × 350)	1,750	\$5/M	\$0.009
Generic-comment detector	2,500 in / 250 out	\$1/M, \$5/M	\$0.004
Per-employee subtotal			\$0.021

Plus per-cycle (per company, not per employee): three-findings summary ~\$0.025 (Sonnet, recommended for this one synthesis call).

Effective per-employee per-cycle: ~\$0.022 with Haiku.

With hybrid (Haiku default + Sonnet for ~15% outlier-review regenerations):

Component	Cost
Base Haiku	\$0.022
15% upgraded to Sonnet at +3× delta	+\$0.005
Per-employee per-cycle (hybrid)	~\$0.027

Per-customer annual cost (4 quarterly cycles)

Company size	Haiku-only	Hybrid (recommended)	All Sonnet
50 employees	\$4.40/yr	\$5.40/yr	\$14/yr
200 employees	\$17.60/yr	\$21.60/yr	\$56/yr
500 employees	\$44/yr	\$54/yr	\$140/yr
1,000 employees	\$88/yr	\$108/yr	\$280/yr
5,000 employees	\$440/yr	\$540/yr	\$1,400/yr

Multiply by ~1.2× safety factor (failed calls, retries, edge regenerations). Real cost ceiling:

- **Small (50 employees): ~\$7/year**
- **Mid (500 employees): ~\$65/year**
- **Large (5,000 employees): ~\$650/year**

Connector costs

For the customer: free. They already pay for Salesforce, BambooHR, GitHub, Slack. Skillnex consumes their existing API allowance.

For Skillnex:

Cost	Pilot	Phase 2	Year 1
Per-customer per-month integration API access	\$0	\$0	\$0
Salesforce AppExchange listing (one-time + annual review)	—	—	~\$5K (optional, only when 3+ Salesforce customers)
Workday Partner program (annual)	—	—	\$20-50K (only if enterprise customer demands Workday)
GitHub Marketplace, Slack App Directory, etc.	\$0	\$0	\$0
Sandbox accounts at every connected provider	\$0	\$0	\$0 (free dev tiers)

The pattern: every modern SaaS exposes a free read-only API for ecosystem partners. We pay nothing per call.

The one expensive partnership is **Workday** at \$20-50K/year — and it pays for itself in months on a single enterprise customer. Skip until enterprise demand justifies it.

Reference market pricing

Product	Per-seat per-month	Annual per-seat
Lattice	\$11	\$132
15Five	\$8	\$96
Culture Amp	\$9-15	\$108-180
Leapsome	\$8-14	\$96-168
ChartHop (HR analytics)	\$4-8	\$48-96
Visier (enterprise people analytics)	\$30+	\$360+

Per product-validation.md §Q2: companies pay \$6-15/user/month for performance management.
Mid-market budget \$50K-\$200K annually for People tech.

Pricing model (locked: per-seat)

Customer	Tier	Monthly	Annual revenue	Direct cost	Gross margin
50 employees	\$9/user/mo	\$450/mo	\$5,400	~\$7 LLM + ~\$50 infra share = \$57	98.9%
500 employees	\$7/user/mo	\$3,500/mo	\$42,000	~\$65 LLM + ~\$200 other = \$265	99.4%
5,000 employees	\$5/user/mo	\$25,000/mo	\$300,000	~\$650 LLM + ~\$500 other = \$1,150	99.6%

LLM cost is genuinely irrelevant. Real costs are people, sales, support.

Pilot pricing

- Free 90-day pilot
- Then \$7/seat/mo month-to-month for 6 months
- Then annual contract

Risk-free try for the law firm. Real reference customer for us.

Defensibility pitch (not efficiency)

Per product-validation.md §Q8: sell defensibility, not efficiency. But efficiency closes deals.

"Your HR team currently spends 3 days per cycle, per HR analyst, manually pulling data for reviews. At a fully-loaded HR analyst cost of \$80,000/year, that's roughly \$1,200 of HR labor per cycle, \$4,800/year, for a 100-person team. Skillnex replaces that prep work for \$7-50/year in API costs. We charge \$5,400/year. The math works the day you turn it on."

Updated business model — full P&L per customer

500-employee mid-market customer (the validation-brief sweet spot):

Line item	Annual
Revenue (500 × \$7/mo × 12)	\$42,000
LLM API (Haiku + Sonnet hybrid)	\$65
Anthropic ZDR uplift (post-pilot)	\$7
Connector APIs (Salesforce, BambooHR, GitHub, etc.)	\$0

Line item	Annual
DO infrastructure (per-customer share at scale)	\$50
Resend transactional email	\$5
Vanta SOC 2 (per-customer share)	\$100
Cyber liability insurance share	\$30
Total direct cost per customer	~\$260
Gross margin per customer	99.4%

2,000-employee enterprise customer with Workday (\$15/seat/mo):

Line item	Annual
Revenue (2,000 × \$15 × 12)	\$360,000
LLM cost	\$260
Workday Partner program (amortized: 4 Workday customers)	\$7,500
Infra	\$200
Other shared (Vanta, insurance, Resend)	\$200
Total direct cost	~\$8,200
Gross margin	97.7%

Workday partner is the biggest single cost line in the entire stack. Pays for itself in months on a single enterprise account.

Cost-monitoring tooling (day one)

```
``sql CREATE TABLE usage_log ( id INTEGER PRIMARY KEY AUTOINCREMENT, tenant_id TEXT NOT NULL, user_id TEXT, model TEXT NOT NULL, -- 'claude-haiku-4-5' | 'claude-sonnet-4-5' input_tokens INTEGER NOT NULL, output_tokens INTEGER NOT NULL, cached_tokens INTEGER NOT NULL DEFAULT 0, cost_usd REAL NOT NULL, endpoint TEXT, -- '/api/analyze', etc. ts TEXT NOT NULL ); CREATE INDEX idx_usage_tenant_ts ON usage_log(tenant_id, ts); ``
```

Captured by middleware on every Anthropic call. Surfaces in `/app/settings/billing`:

- Tokens this cycle, projected monthly cost
- Per-model breakdown
- Cache hit rate (margin signal)
- Internal alert when single customer crosses \$100/cycle (something is wrong)

What scales cheaply, what doesn't

Stays cheap:

- Adding a 6th, 7th, 8th competency per employee → linear scale, still cents
- Adding integration-pulled signals to the prompt → +20% input tokens, still cents
- Adding the generic-comment detector → already costed
- Per-employee Sonnet upgrade button → costed at 15% rate
- Cycle-close three-findings in Sonnet → single-digit cents per company

Gets expensive (would change pricing):

- Real-time narrative regeneration on every signal pull (vs at cycle close): 10× call volume. Stay scheduled, not real-time.
- Embedding-based "find similar employees": ~\$0.50/employee/year additional. Phase 3.
- "Chat with your data" multi-turn: 10-30× per-employee cost. Phase 3+ enterprise tier.
- Auto-running narratives on every cycle for every employee (vs on-demand): 100,000 calls/year for 5,000-employee company. Still ~\$1,000/year on Haiku, \$3,000 on Sonnet. Affordable but worth opt-in.

Build sequence (3 days, slots into Phase 2)

Day	Work
1	usage_log table + token-tracking middleware on /api/analyze. Backfill existing narratives.
1	Cost dashboard in /app/settings/billing (admin-visible only) — tokens this cycle, projected monthly, model breakdown, cache hit rate.
2	Model router (lib/llm/router.ts) with Haiku-default + Sonnet-upgrade rules.
2	"Regenerate with deeper analysis" button on NarrativeCard — calls API with regenerate_with_deeper_analysis: true, uses Sonnet, marked clearly in UI.
3	Three-findings cycle-close synthesis endpoint (Sonnet).
3	Anthropic Prompt Caching: confirm cache writes happening across per-competency calls. Verify cache_read_input_tokens > 0 in usage logs.
3	Tests: model router selects correctly given employee + flags + request body. Usage log captures every call.

Tests

Test	Verifies
<code>tests/llm/router.test.ts</code> — Director-level employee → Sonnet	Auto-upgrade for senior
<code>tests/llm/router.test.ts</code> — score-vs-rating-mismatch flag → Sonnet	Auto-upgrade for outliers
<code>tests/llm/router.test.ts</code> — <code>regenerate_with_deeper_analysis: true</code> → Sonnet	On-demand upgrade
<code>tests/llm/router.test.ts</code> — default employee + no flags + no regenerate → Haiku	Default behavior
<code>tests/llm/usage-log.test.ts</code> — every analyze call writes a usage row	Cost tracking
<code>tests/llm/usage-log.test.ts</code> — <code>cache_read_input_tokens</code> recorded	Cache verification
<code>tests/llm/three-findings.test.ts</code> — cycle-close synthesis uses Sonnet, one call per cycle	Sonnet routing

Items deferred

- Embedding-based similar-employee search — Phase 3
- Multi-turn "chat with your data" — Phase 3 enterprise
- Per-customer LLM budget cap — never (decision #7)
- Freemium tier — month 6, only if needed for growth
- Cost-passed-through pricing tier — never (we eat ZDR uplift, decision #8)